Name:		Centre/Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY — 9744/02

Paper 2 Structured Questions

13 September 2021

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your centre number, index number, name and class at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the question paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	10
2	8
3	13
4	10
5	10
6	10
7	10
8	9
9	7
10	8
11	5
Total	100

This document consists of **28** printed pages (including this cover page) and **4** blank pages.

For Examiner's use

Answer **all** questions in this section.

Question 1

Fig 1.1 shows the structure of a mammalian eye. A section of the retina is boxed up for a close up look.

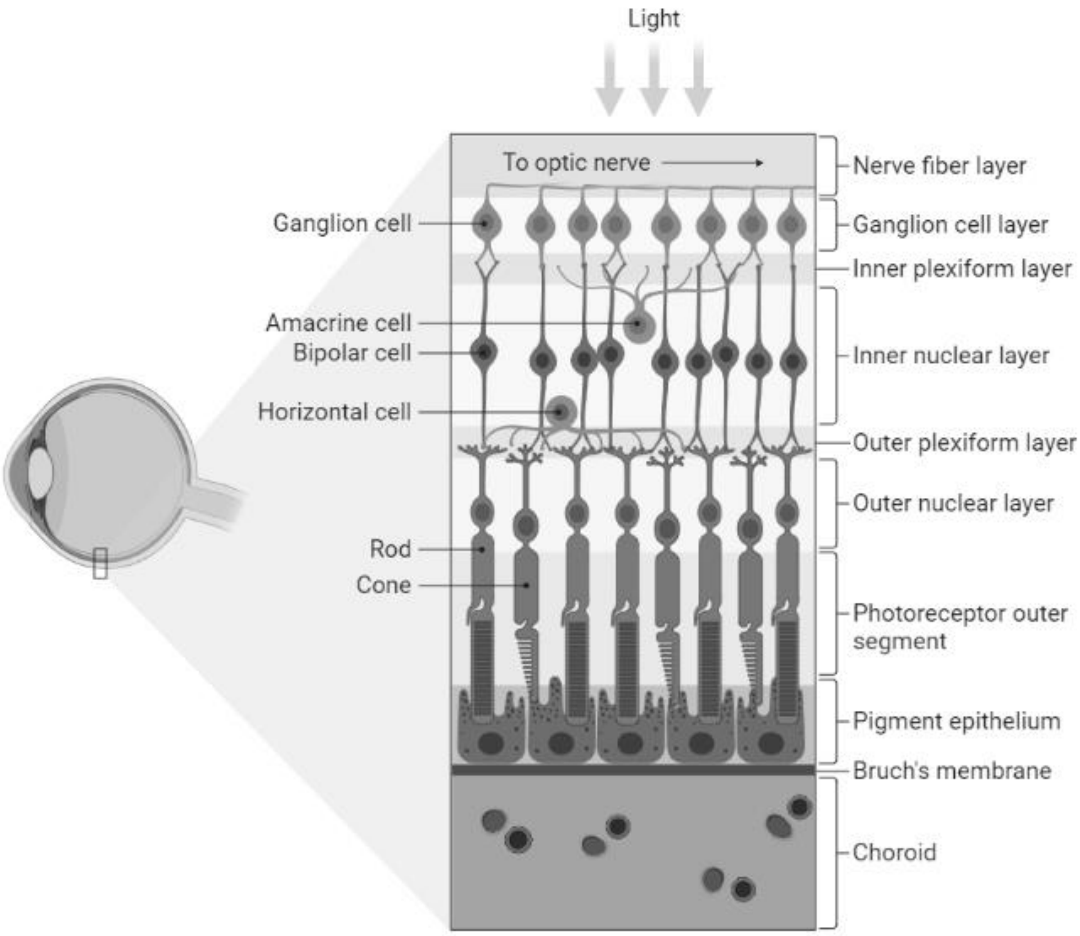

Fig 1.1

(a) Explain how Fig 1.1 supports the cell theory.

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.... [2]

Fig 1.2 shows an electron micrograph of part of a rod cell, one of the photoreceptor cells in the retina.

For Examiner's use

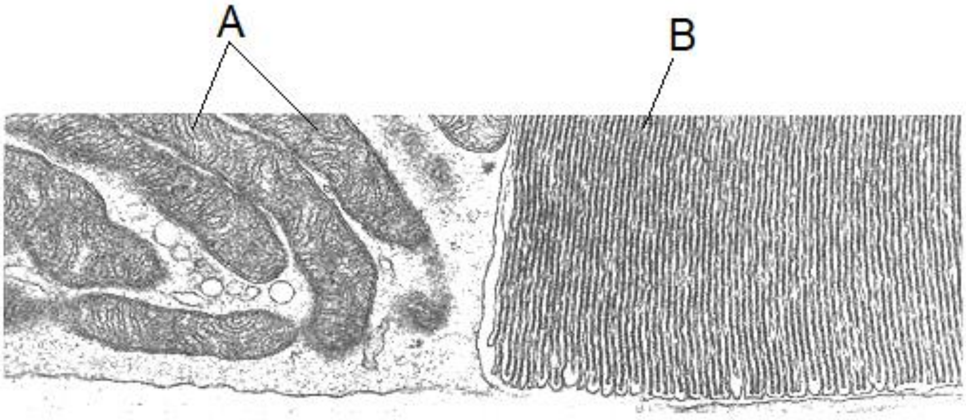

Fig 1.2

(b) **(i)** With reference to Fig 1.2, identify organelle **A**.

.... [1]

(ii) Using your knowledge, explain how organelle **A** is suited for its function in a cell.

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.... [3]

For Examiner's use

(c) Fig 1.3 shows a drawing of a rod cell and an organelle found in plant cells. Structure **B** can also be seen in Fig 1.2.

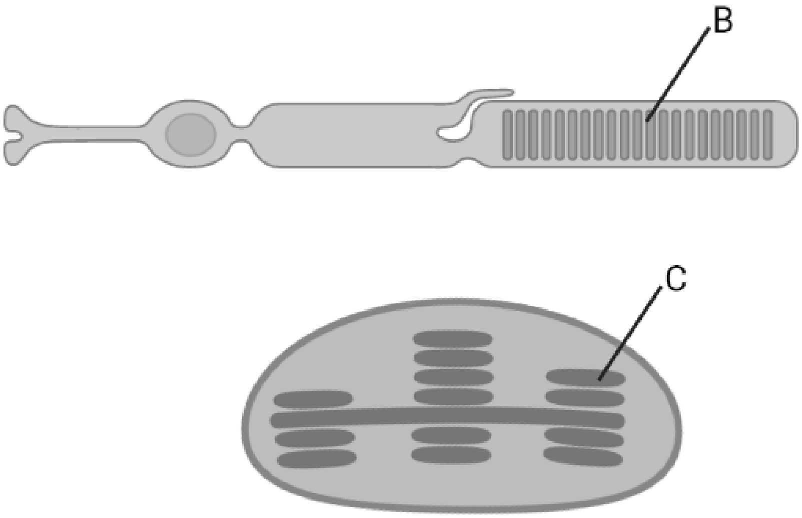

Fig 1.3

Structure **B** is a disc found in the outer segment of the rod cell. The disc is a membrane-enclosed sac packed with photo-sensitive molecules known as *rhodopsin*.

Identify structure **C** and compare structures **B** and **C**.

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.... [4]

[Total: 10]

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For Examiner's use

Question 2

Fig 2 shows the synthesis of collagen.

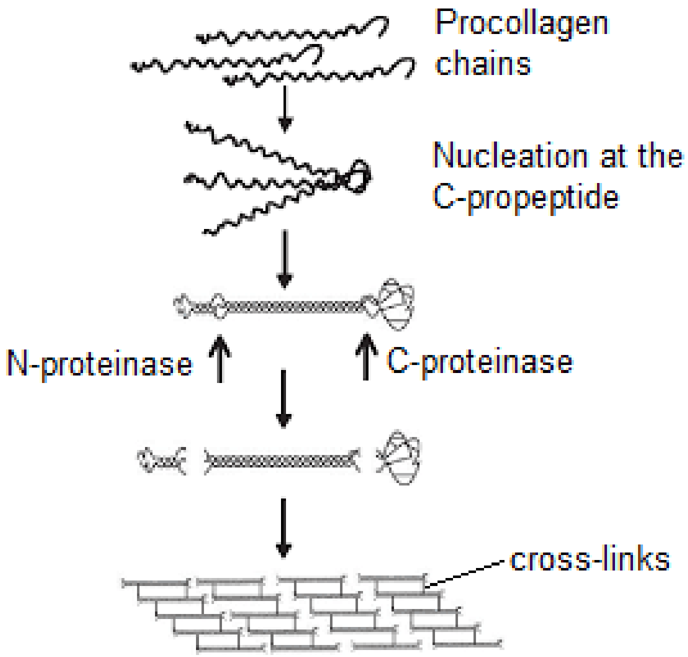

Fig 2

(a) **(i)** In Fig 2, box out and label tropocollagen. [1]

(ii) With reference to Fig 2 and using your own knowledge, describe the synthesis of tropocollagen from procollagen chains.

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.... [3]

For Examiner's use

(b) **(i)** One of the covalent cross-links in collagen is labeled in Fig 2.

Using your knowledge, deduce whether the covalent cross-link is formed inside or outside the cell.

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.... [3]

(ii) Covalent cross-links contributes to a property of collagen which is important for its function.

Identify another property of collagen which is important for collagen's function.

.... [1]

[Total: 8]

For Examiner's use

Question 3

Fig 3 shows a section of the thylakoid membrane.

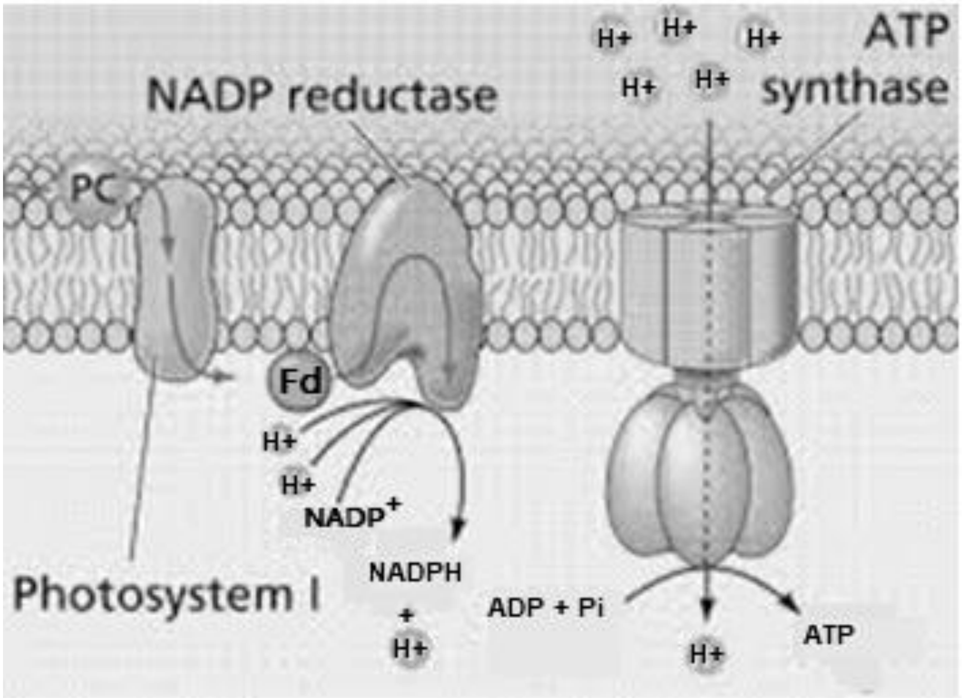

Fig 3

(a) Explain how the membrane in Fig 3 supports the fluid mosaic model.

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.... [3]

(b) Describe how H^+ is transported across ATP synthase.

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.... [1]

For Examiner's use

(c) Explain how NADP reductase is stabilised in the thylakoid membrane.

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.... [2]

(d) Briefly explain the mode of action of the enzyme NADP reductase.

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.... [3]

(e) With reference to Fig 3, explain how an increase in temperature increases the rate of photosynthesis.

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.... [4]

[Total: 13]

For Examiner's use

Question 4

Fig 4a shows a replication fork that is opening towards the right during DNA replication. The replication fork opens up till one end of the chromosome is reached. The parental DNA strands are shown in black with the 5' and 3' ends labelled. Daughter DNA synthesis is shown as grey arrows. Fig 4b shows the parental end of chromosome.

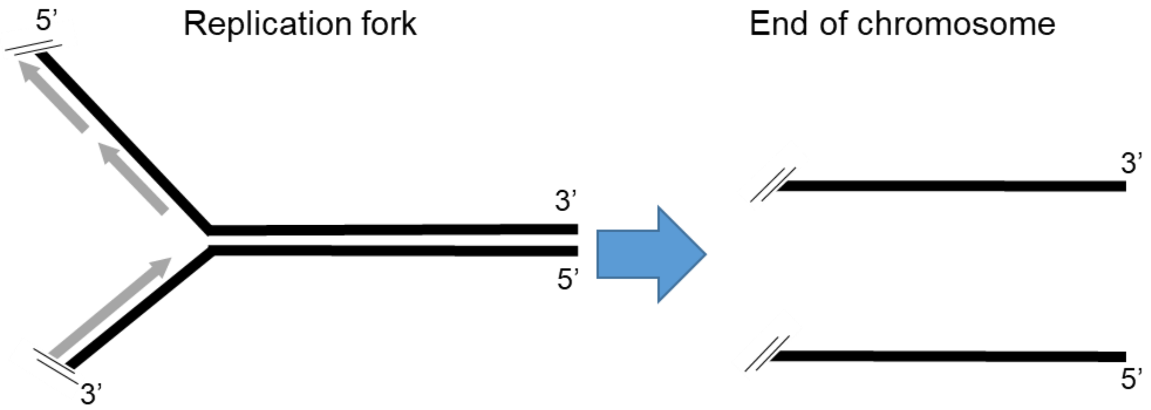

Fig 4a **Fig 4b**

(a) State the name of the enzyme responsible for opening the replication fork.

.... [1]

(b) On Fig 4a, label the leading and lagging strands on the replication fork. [1]

(c) Explain why DNA replication requires both leading and lagging strand synthesis.

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.... [2]

(d) Draw on Fig 4b the newly synthesised daughter strands of DNA after DNA replication is completed. [2]

For Examiner's use

(e) Compare DNA polymerase III and Telomerase.

[4]

[Total: 10]

For Examiner's use

Question 5

Scientists have found a way to insert the human gene for insulin into bacteria. These recombinant bacteria can then be used to produce insulin for medical purposes. The gene for insulin is first inserted into a plasmid. Fig 5.1 shows how the plasmid is prepared.

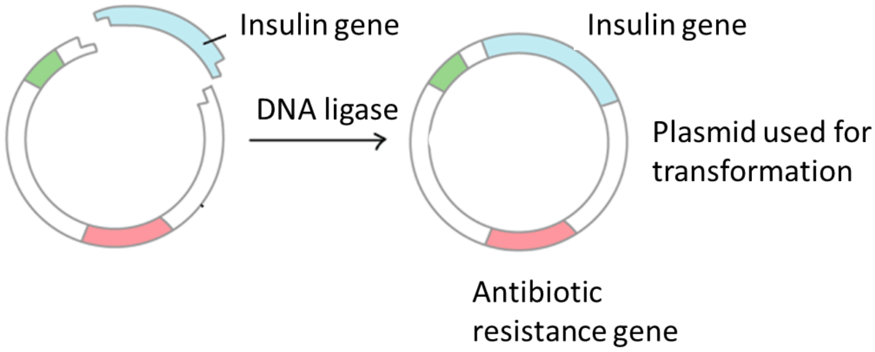

Fig 5.1

The plasmid is then inserted into the bacteria by heat shock treatment which creates transient pores in the bacterial cell membrane. The bacteria are then grown on agar with antibiotic. Colonies that grow are then used in culture to create insulin. Fig 5.2 shows a summary of this process.

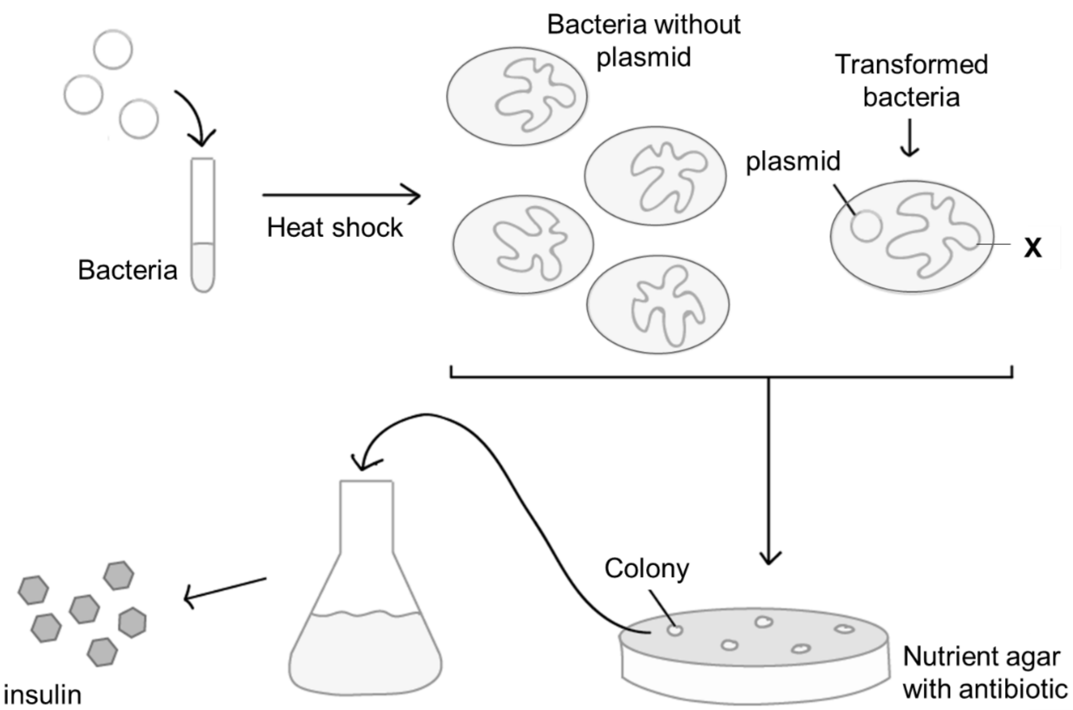

Fig 5.2

For Examiner's use

(a) With reference to Fig 5.1, explain the function of DNA ligase.

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.... [2]

(b) With reference to Fig 5.2, identify molecule X.

.... [1]

(c) Describe how this method of artificial bacterial transformation differs from natural bacterial transformation.

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.... [2]

(d) Suggest why bacteria are grown on an agar plate with antibiotic after the heat shock treatment.

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.... [2]

For Examiner's use

(e) Each bacterial colony on the agar plate is made up of genetically identical clones formed from a single bacterium that is able to produce insulin.

Explain how a single bacterium is able to form such a colony.

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.... [3]

[Total: 10]

For Examiner's use

Question 6

Fig 6.1 shows microscopy images (**A** to **I**) of different stages in meiosis.

The images are not shown in order of stages in meiosis.

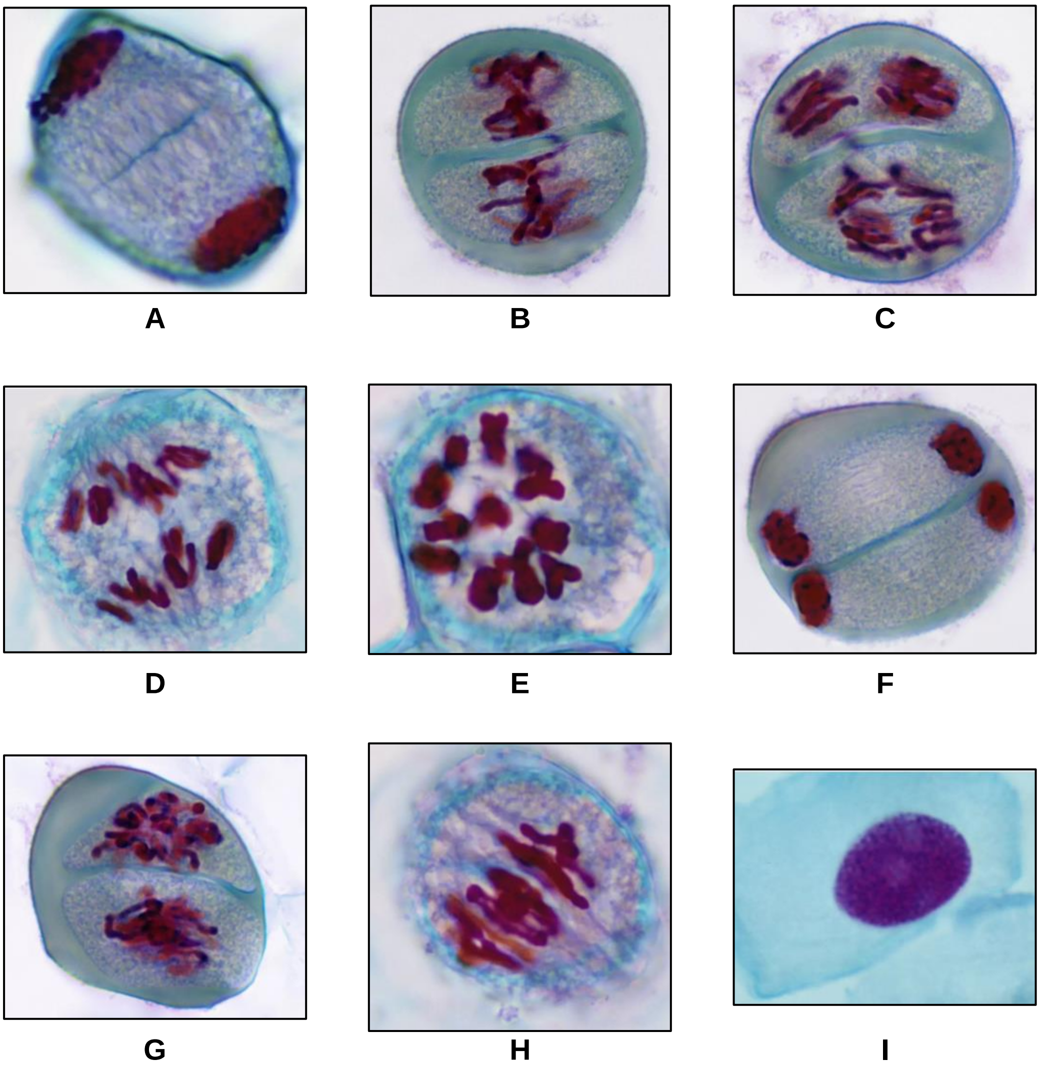

Fig 6.1

(a) With reference to Fig 6.1, describe two differences between stage B and stage H.

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.... [2]

Fig 6.2 shows the arrangement of a pair of homologous chromosomes (**P** and **Q**) during one of the stages in the meiotic cell cycle.

For Examiner's use

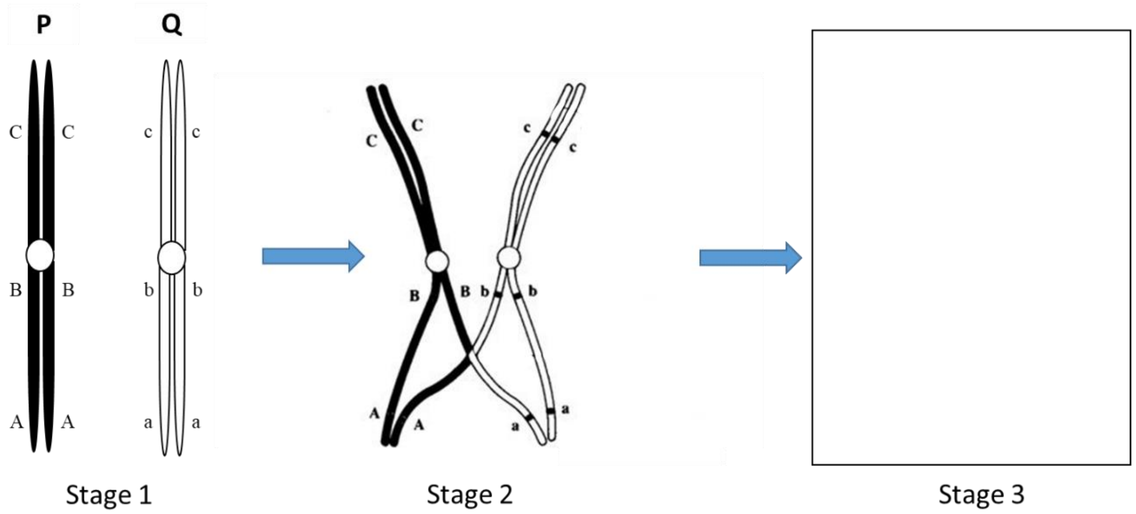

Fig 6.2

(b) With reference to Fig 6.2,

(i) explain how structure **P** is different from structure **Q**,

....

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.... [2]

(ii) identify the stage (**A** to **I**) in Fig 6.1 that contains the arrangement of chromosomes observed in stage 2,

.... [1]

For Examiner's use

(iii) describe what is occurring in stage 2 and explain the significance of this event in meiosis,

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.... [4]

(iv) draw in the box provided, the structure of the chromosomes in stage 3. Include the allele labels in your drawing. [1]

[Total: 10]

For Examiner's use

Question 7

In fruit flies, *Drosophilia melanogaster*, a recessive mutant allele (**r**) leads to purple eyes as opposed to the normal allele (**R**) for red eyes. A second recessive mutant allele (**n**) leads to vestigial wings as opposed to normal allele (**N**) for normal length wings.

(a) **(i)** Suggest why mutations are usually recessive.

[2]

(ii) Explain how recessive alleles are preserved in a natural population.

[2]

For Examiner's use

A test cross was conducted on a heterozygote fly resulting in the following offspring:

1139 red eye, normal wing flies
1195 purple eye, vestigial wing
51 red eye, vestigial wing
55 purple eye, normal wing

(b) Draw a genetic diagram to explain the observed results.

[4]

For Examiner's use

The phenotype of an individual is determined by its genotype. However, the environment may influence gene expression.

(c) Explain how temperature affects the fur colour of Himalayan rabbits.

[2]

[Total: 10]

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For Examiner's use

Question 8

Fig 8 shows the signal transduction pathway in muscle cells in response to the hormone, epinephrine.

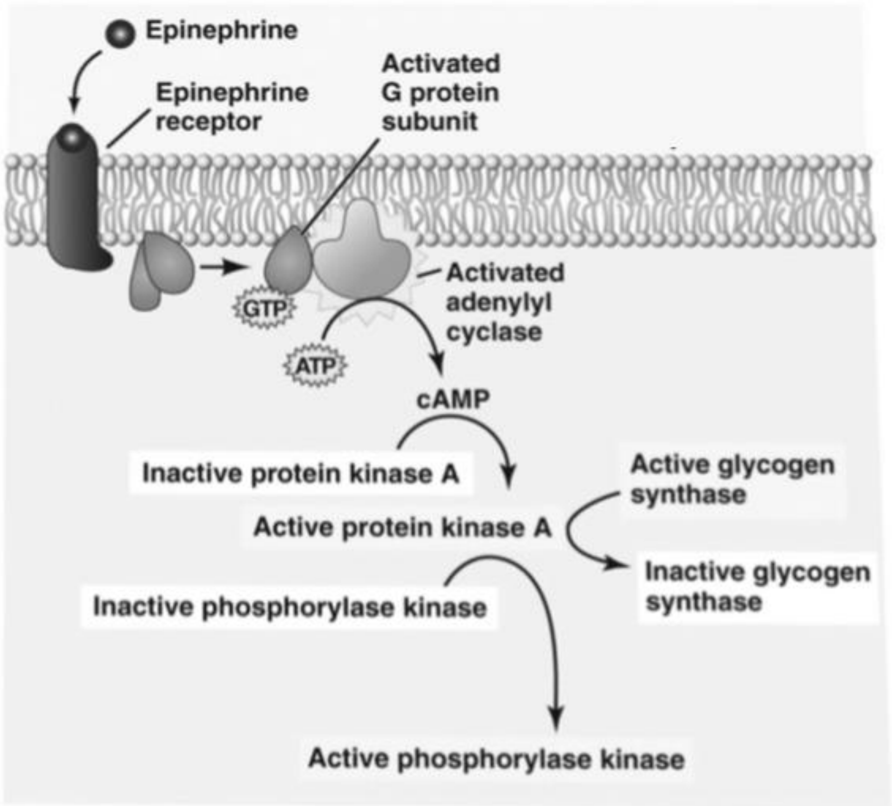

Fig 8

(a) Explain how the structure of the epinephrine receptor allows it to carry out its function.

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.... [3]

Various drugs can be designed to alter the response of muscle cells to epinephrine. It was found that a drug inhibits the GTPase activity.

For Examiner's use

(b) With reference to Fig 8, suggest the effect of the above drug on muscle cell.

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.... [4]

(c) Explain how epinephrine can have different effects on different cells.

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.... [2]

[Total: 9]

Question 9

For Examiner's use

Evolutionary changes in the beaks of Darwin's finches have been instrumental in the adaptive radiation of these birds in the Galapagos Islands.

There are a dozen species that are closely related to one another showing remarkable differences in beak shape and size, and in eating habits. They have all arisen from a common ancestor, but have evolved to exploit different food sources.

The precise dimensions (length, depth and width) of each species' beak are crucial to their survival, and fluctuations in the environment lead to selection that changes the relative success of birds with various beak shapes.

The molecular basis for variation in beak size and shape is due to two genes:

the gene for **bone morphogenetic protein 4 (BMP4)** that encodes a protein which allows for cell-to-cell communication that is expressed more during the embryonic development of the deep and wide beaks of ground finches than during the narrower beaks.

the gene for **calmodulin (CaM)** that encodes a calcium signalling protein that is expressed at much higher levels in the long and pointed beaks of cactus finch embryos.

Fig **9** shows how these two genes interact to produce variation in beak size and shape in the different finch species.

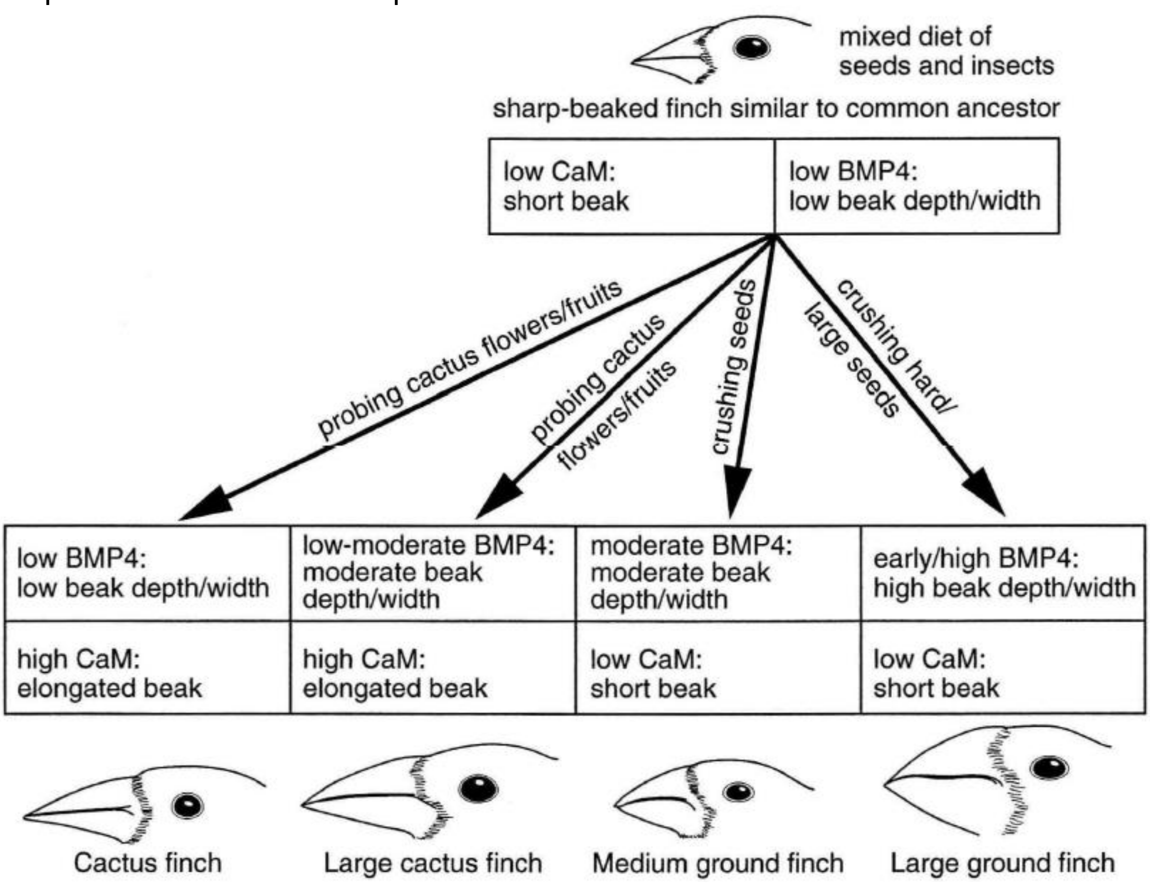

Fig 9

For Examiner's use

(a) Explain how the difference in beak size and shape between species may have arisen.

[5]

(b) Explain why a population is the smallest unit of evolution.

[2]

[Total: 7]

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For Examiner's use

Question 10

Tuberculosis (TB) is caused by the bacteria *Mycobacterium tuberculosis*. The most common first line medications used to treat tuberculosis include Isoniazid and Rifampin. Table 10 shows the mode of action of these 2 drugs.

Table 10

First line drug	Mechanism of action
Isoniazid	Inhibits cell wall synthesis
Rifampin	Inhibits RNA synthesis

(a) With reference to Table 10, explain how Isoniazid and Rifampin are able to kill the bacteria *Mycobacterium tuberculosis.*

[2]

Fig 10 shows the percentages of patients in New York City who were infected with TB resistant to one or more antituberculosis medications between the years 1953 to 1991.

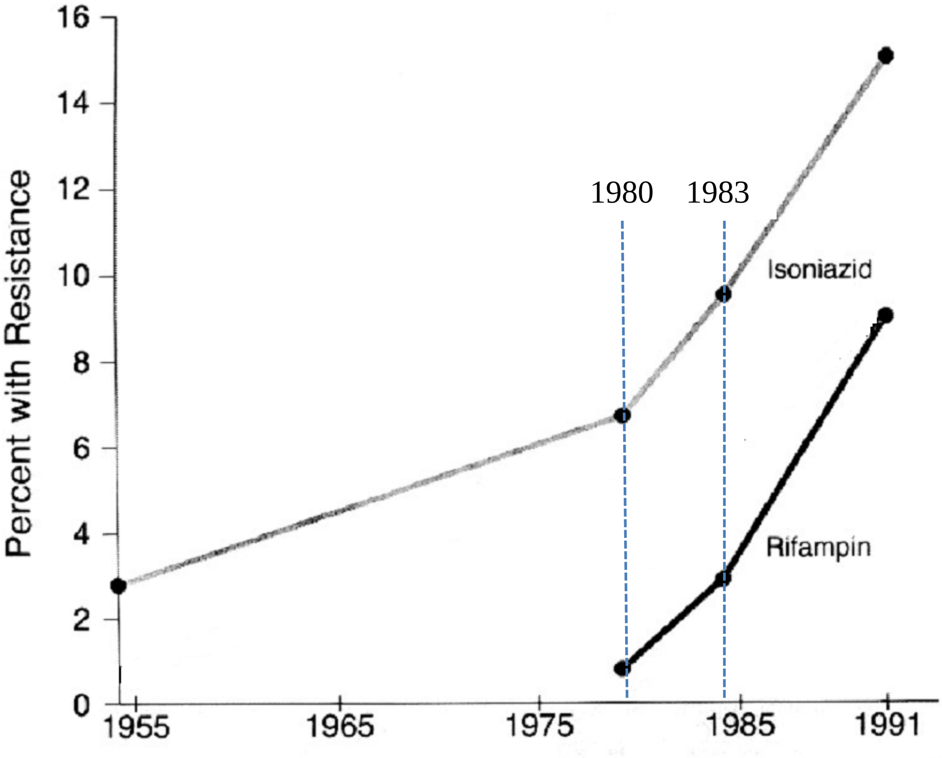

Fig 10

(b) With reference to Fig 10, compare the changes in percentage of patients with TB resistant to Isoniazid and Rifampin.

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.... [2]

For Examiner's use

For Examiner's use

(c) Deduce which drug, Isoniazid or Rifampin, was first to be used in treatment of TB and explain your reasoning.

[2]

People with *compromised immune systems*, such as people living with HIV, malnutrition or diabetes, or people who use tobacco, have a higher risk of falling ill with an active TB infection.

(d) Explain how having a HIV infection can result in an individual having a "*compromised immune system*".

[2]

[Total: 8]

Question 11

For Examiner's use

Over thousands and millions of years, the Earth's climate changes with periods of warming and cooling. Fig. 11 is a diagram showing the topographical profile of two mountains in the tropics during a warm phase and a cool phase in the Earth's climate. The shape of the lines corresponds to a vertical section through the mountains to show their height and shape. The distribution of rain forest vegetation is also shown.

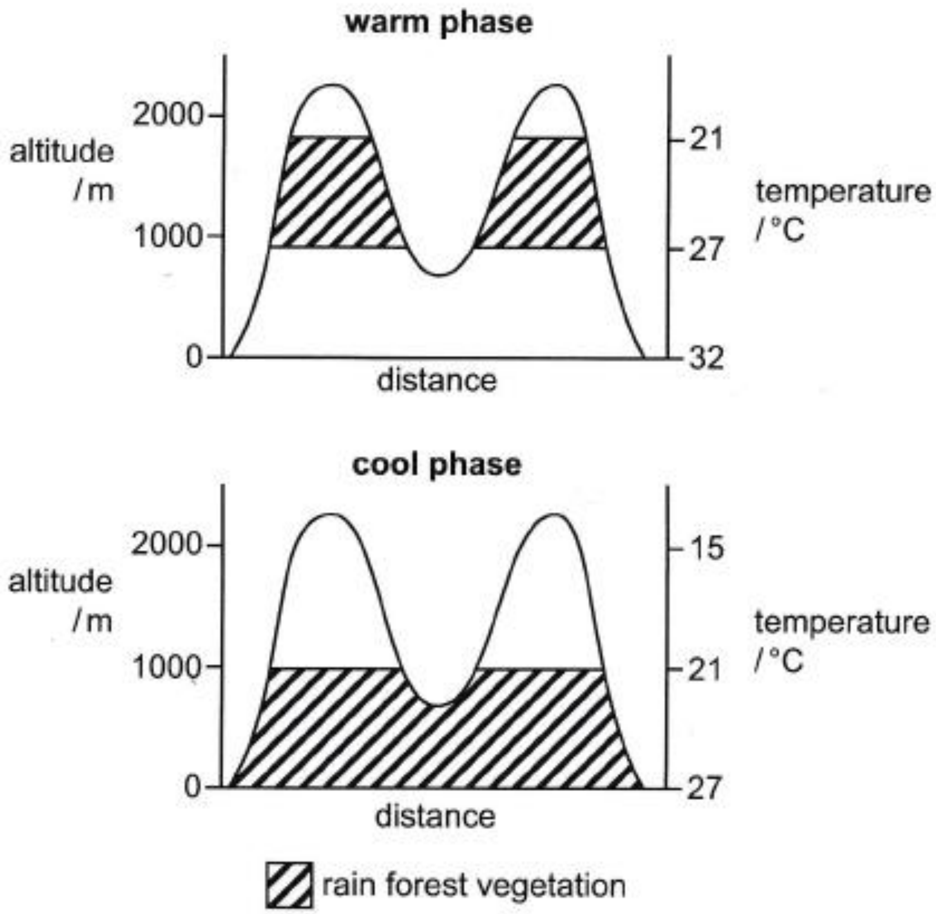

Fig. 11

Describe and explain the effect of climate change on the distribution of rain forest vegetation in the tropics, as shown in Fig.11.

For Examiner's use

[5]

[Total: 5]

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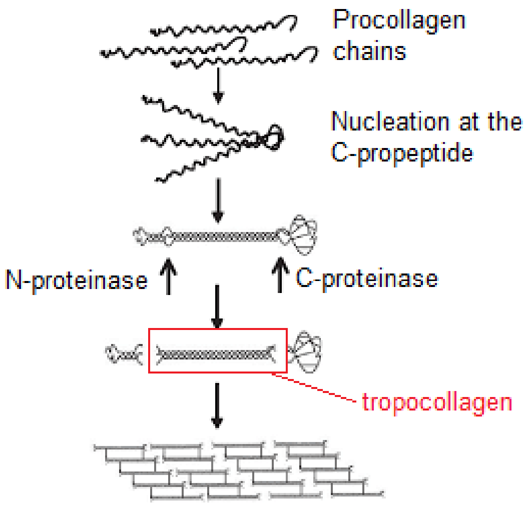
DUNMAN HIGH SCHOOL
DHP YR 6
H2 BIOLOGY

Year 6 H2 Preliminary Exam 2021
Suggested Answers

Paper 2

Question 1			
(a)	The retina layer is composed of multiple cell types + name at least 2 cell types; Supporting the cell theory that living organisms are made composed of;		[2]
(b)	(i)	Mitochondria;	[1]
	(ii)	Highly folded inner membrane / cristae; Enables many electron carriers / electron transport chains / ATP synthase to be embedded; For high rate of ATP synthesis; For cellular activities; Max 3	[3]
(c)		Thylakoid; Any 3 below - Both are single membrane bound; Both contains many photo-sensitive molecules; B is located within the cytoplasm of the cell while C is located within a chloroplast in a cell; B is found in animal cells while C is found in plant cells;	[4]

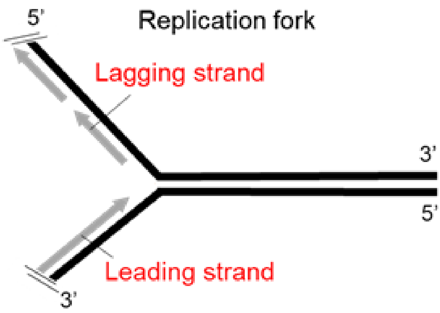
Question 2			
(a)	(i)	 Correctly box out and label tropocollagen;	[1]
	(ii)	3 Procollagen chains nucleate at C-terminus / C-propeptide; Hydrogen bonds formed between C=O of proline on one chain and N-H of glycine on another chain; Stabilising the triple helix; Ends cleaved by <u>N-proteinase</u> and <u>C-proteinase</u>; Any 3	[3]
(b)	(i)	Formed outside; Any 2 from - Covalent cross-links between tropocollagen form fibrils; Fibrils are large ; If they are synthesized in the cell, the fibrils will not be able to pass through the cell surface membrane into the extracellular matrix / tendons where they function;	[3]
	(ii)	Insoluble;	[1]

Question 3		
(a)	Random arrangement of proteins embedded in membrane; Proteins and lipids able to move rapidly and laterally in the plane of the membrane; ref to NADP reductase and ATP synthase embedded; OR Ref to PC able to move towards photosystem I;	[3]
(b)	Facilitated diffusion down concentration gradient from thylakoid space to stroma;	[1]
(c)	Globular structure with hydrophobic regions interacting with hydrophobic core of the phospholipid bilayer; Hydrophilic regions facing the aqueous thylakoid space and stroma; OR Hydrophilic regions interacting with polar phosphate head of phospholipid;	[2]
(d)	<u>Active site</u> specific / complementary; to any one substrate - Fd / electron, H^+ or $NADP^+$; Form enzyme-substrate complex + reduces the <u>activation energy</u>; Catalyses the reduction of $NADP^+$ with the addition of H^+ and electron; Any 3	[3]
(e)	Increase kinetic energy of enzyme and substrate; Ref to one enzyme and respective substrate; Eg NADP reductase and $NADP^+$ / electron / H^+ / Fd OR ATP synthase and ADP / Pi Increase rate of effective collision; Increase in rate of enzyme-substrate complex formation;	[4]

Question 4

(a) Helicase; [1]

(b) [1]

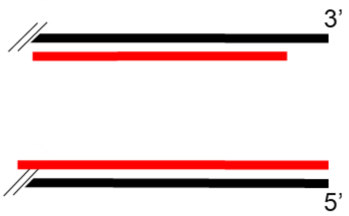


leading strand at the bottom and lagging strand on the top;

(c) [2]

DNA is antiparallel;
DNA polymerase III can only synthesise DNA in the 5' to 3' direction;
Leading strand synthesized continuously + lagging strand synthesized discontinuously in Okazaki fragments;
Max 2

(d) [2]



Lagging strand drawn to end short, not reaching the 3' end of the parental template;
Leading strand drawn to reach 5' end of parental template;

(e) [4]

Similarities (3max)
Both synthesize DNA;
complementary base pairing;
synthesis in 5' to 3' direction;
AVP;

Differences (3 max)

	DNA pol III	Telomerase
Part of DNA that is replicated	Replication of entire length of chromosome	Extends the 3' end of parental template
Template	No internal template, uses parental DNA as a template	Has internal RNA template
Base pairing	Complementary base pairing for synthesis is between dNTPs and DNA	Complementary base pairing for synthesis is between dNTPs and RNA
presence	In all replicating cells	In stem cells only

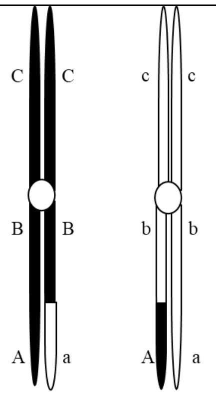
AVP;

Total 4 max

<table>
<tr><td colspan="3">Question 5</td></tr>
<tr><td>(a)</td><td>Catalyse the formation of phosphodiester bonds;
To join the insulin gene with the plasmid;</td><td>[2]</td></tr>
<tr><td>(b)</td><td>Bacterial chromosome;</td><td>[1]</td></tr>
<tr><td>(c)</td><td>
<table>
<tr><td></td><td>artificial bacterial transformation</td><td>natural bacterial transformation</td></tr>
<tr><td>Type of DNA</td><td>Only plasmid is used</td><td>Fragment of linear DNA or plasmid can both be used</td></tr>
<tr><td>How DNA is taken up</td><td>Heat shock to create pores</td><td>DNA binding protein on cell surface</td></tr>
<tr><td>Recombinase / Rec A</td><td>No need for recombinase / rec A</td><td>Need for recombinase / rec A to incorporate DNA into bacterial chromosome</td></tr>
</table>

2 max</td><td>[2]</td></tr>
<tr><td>(d)</td><td>Not all bacteria are successfully transformed after the heat shock treatment;
Selects for bacteria that have successfully taken up the plasmid and thus have the antibiotic resistance gene;</td><td>[2]</td></tr>
<tr><td>(e)</td><td>Semiconservative DNA replication of the bacterial <u>chromosome</u>;
<u>plasmid</u> with insulin gene is self-replicating;
Bacterial chromosome and plasmid are attached to the mesosome / cell membrane as cell elongates, pulling them apart;
Cell divides into two by <u>binary fission</u>, forming 2 identical cells with the plasmid inside;

3 max</td><td>[3]</td></tr>
<tr><td></td><td></td><td></td></tr>
</table>

Question 6			
(a)	1. Homologous pairs of chromosomes align at metaphase plate in stage H whereas single chromosomes with 2 sister chromatids align at the metaphase plate in stage B. 2. Cell in stage H has double the amount of DNA in comparison to each cell in stage B. OR Cell in stage H has double the number of chromosomes in comparison to each cell in stage B. 3. Stage B – 2 microtubule attached to each chromosome, Stage H – 1 microtubule attached to one chromosome. 2 max		[2]
(b)	(i)	Different alleles of the genes, P has alleles A, B and C while Q has a, b and c; Different origins – one maternal and one paternal;	[2]
	(ii)	E;	[1]
	(iii)	<u>Crossing over</u>; occurs where non-sister chromatids of homologous chromosomes break, exchange and re-join; give rise to recombination of alleles, Abc on one chromatid and aBC on the other chromatid; OR allows separation of linked genes in gametes formed; contributes to genetic diversity in gametes formed;	[4]
	(iv)	 center chromatids showing exchange of chromatid arms between B/b and A/a loci;	[1]

Question 7

(a)	**(i)**	Change in amino acid sequence + change in 3D conformation of protein + loss of function; The level of expression of one wild type allele (dominant) is sufficient to produce the wild type phenotype / mask the mutant allele (recessive);	[2]
	(ii)	Individuals are <u>diploid</u> and genetic variation can be maintained in the form of recessive alleles in heterozygotes, allowing recessive alleles to persist in the population.; There may be balancing selection / frequency-dependent selection such that the individuals that are too common declines as they are targeted by predators, preserving polymorphism.; There is variation in environment / habitats, allowing individuals with different genotypes and phenotypes to survive and reproduce, preserving polymorphism.; Due to <u>heterozygote advantage</u>, recessive alleles can be preserved in heterozygotes.; There is neutral selection / genetic variation that confer no selective advantage, maintaining the frequency of recessive alleles in population.; <u>Max 2</u>	[2]
(b)		See genetic diagram below	[4]

Parental phenotypes:	Red eye, normal wing	X	Purple eye, vestigial wing
Parental genotype:	R N r n	X	r n r n
Gametes formed:	R N, r n R n, r N	X	r n

F$_1$ genotypes and phenotypes:

	R N	r n	R n	r N
r n	R N r n	r n r n	R n r n	r N r n
	Red eyes, normal wings	Purple eyes, vestigial wings	Red eyes, vestigial wings	Purple eyes, normal wings

Parental phenotypes (higher proportion)

Recombinant phenotypes (lower proportion)

	1m – parental phenotype; 1m – genotypes (linked); 1m – gametes (circled); 1m – F1 genotypes and phenotypes;	
(c)	Heat from environment prevents the development of black fur.; Tyrosinase required for melanin production is heat sensitive and is denatured at higher temperatures. Hence, fur-producing cells do not produce melanin and fur appears white.;	[2]

Question 8		
(a)	Epinephrine receptor is a G-protein coupled receptor (GPCR) with 7 transmembrane alpha helices.; Extracellular binding site has a specific conformation that is complementary to epinephrine in terms of shape and charges.; Intracellular domain can undergo a conformational change upon binding of epinephrine, resulting in binding and activation of G protein.;	[3]
(b)	GTP cannot be hydrolysed to GDP, hence G protein remains constantly active.; α-subunit of G-protein remains bound to adenylyl cyclase, resulting in continuous conversion of ATP to cAMP.; cAMP continuously activates protein kinase A (PKA).; Activated PKA constantly phosphorylate and inactivates glycogen synthase; Activated PKA constantly phosphorylate and activate phosphorylase kinase.; Max 3 above Release / accumulation of large pool of glucose molecules.;	[4]
(c)	Due to presence of different relay molecules / second messengers; activating different transduction pathways and thus cellular responses.;	[2]

Question 9		
(a)	There is variation in beak shape and size in the population of common ancestor / sharp-beaked finch; The variation is due to different expression levels in *CaM* and *BMP4* genes + as a result of random gene mutations and sexual reproduction; Different food sources / availability of food acts as a selection pressure for depth, width and length of beaks of finches.; Ref to any example of food source and beak shape selected; These finches were able to survive and reproduce, passing down alleles for desried traits to their offspring, increasing in frequency over time.; Over a long period of time, the accumulation of multiple changes in genetic makeup (including *BAMP4* and *CaM*) results in the formation of cactus finch, a new species.; Max 5	[5]
(b)	A population can have genetic variation in traits for selection pressure to act on, but an individual cannot acquire variation.; Evolution involves changes in allele frequency in a population of organisms over time.;	[2]

Question 10		
(a)	Isoniazid inhibits synthesis of mycolic acid cell wall, resulting in new cells being fragile and susceptible to osmotic lysis; Rifampin inhibits RNA synthesis, preventing the bacteria from synthesizing essential proteins and enzymes, causing cell death;	[2]
(b)	Both Isoniazid and Rifampin saw a sharp rise in percentage of patients with resistance between 1980 to 1991, (6 to 15% for Isoniazid and 1-9% for Rifampin); Percentage of patients with resistance to Isoniazid is consistently higher than percentage of patients with resistance to Rifampin (quote 1 set of data); OR No resistance against Rifampin observed before 1980 whereas resistance against Isoniazid was already observed in 1953 in 3% of patients.	[2]
(c)	Isoniazid; As resistance for TB for Isoniazid was observed in patients earlier than Rifampin, indicating that Isoniazid was already present as a selection pressure, selecting for bacteria with mutations that gave them resistance to Isoniazid;	[2]
(d)	HIV infects T cells causing their numbers to decrease; Helper T cells required to activate B cell and $CD8^+$ T cells of adaptive immune system;	[2]

Question 11	
<u>Describe</u> (Must quote altitude values for both warm and cool phase) During the warm phase, the distribution of the rainforest vegetation ranges from 900 to 1800m above sea level. During the cool phase, the distribution of the rainforest vegetation ranges from 0 to 1000m above sea level; The distribution of vegetation is continuous in the cool phase, whereas the distribution of vegetation is patchy in the warm phase; <u>Explain</u> As temperature increases, the distribution for rainforest vegetation moves upwards; Rainforest vegetation will only grow in temperature range of 21 – 27°C (must quote values); Ref. to lower temperatures at higher altitudes;	[5]